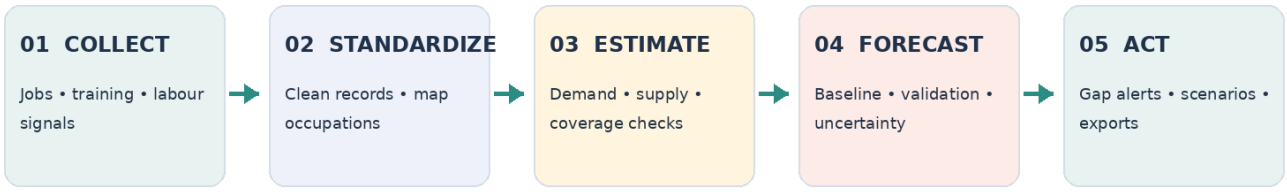
SMART INDIA HACKATHON 2026

MASTER R&D

AI-Enabled Labour Market Intelligence & Skill Demand–Supply Forecasting Engine

Problem understanding • research • data strategy • architecture • build plan • presentation blueprint



Every result should show: value + source + last updated + confidence/limitation.

One consolidated source of truth for the team and any AI used to build the final SIH presentation.

Important: this is a research and implementation blueprint, not proof of live integrations or validated nationwide forecasts.

How to use this Master R&D

This document consolidates the full LMIS build guide and maps the content to the six-slide structure in the uploaded reference deck. Read Sections 1–8 to understand the problem and data; Sections 9–18 for modelling and implementation; Sections 19–23 for demo, risks and sources; Sections 24–25 to reproduce the reference presentation structure.

Reference deck structure observed

- Slide 1 — SIH cover: PS ID/title, theme/category and team metadata.
- Slide 2 — Product overview: current problems on the left, solution capabilities on the right.
- Slide 3 — Technical approach: technologies, central architecture and supporting service blocks.
- Slide 4 — Workflow: numbered journey with system interactions and data movement.
- Slide 5 — Feasibility & viability: technical feasibility, risks/controls and operational viability.
- Slide 6 — Impact & benefits plus research references.

The uploaded reference is about GovSync and government-platform interoperability. Its six-slide structure is a template only; its project claims, stack, credentials and technical details must not be copied into the LMIS project.

1. Executive summary

India trains people across many occupations, but training seats, course choices and local employer needs do not always line up. This project proposes a Labour Market Intelligence System (LMIS): a decision-support platform that combines labour-market signals with training information to estimate where skill shortages or potential oversupply may emerge.

The platform should help planners answer three questions: What work is likely to be needed? Is enough suitably skilled supply likely to be available? What training decision should be examined now, before a mismatch becomes worse?

Core principle
Every result must show the estimate, the evidence behind it, the last-updated date and the limitation/uncertainty. The system supports planning; it does not automatically make government policy decisions.

2. Official problem statement and what it requires

Problem Statement: “AI-Enabled Labour Market Intelligence and Skill Demand-Supply Forecasting Engine.” The brief describes fragmented labour-market and skilling information, mismatches between training capacity and industry demand, and the need for periodic forecasts at sector/trade/district level.

- Aggregate and normalize demand signals from job portals, industry hiring data, NCO/NSQF-coded postings and government employment databases where access is available.
- Cross-reference demand with training capacity and annual seat allocations by sector, trade and district.
- Forecast demand–supply gaps periodically and rank oversupply/undersupply severity.
- Provide an interactive dashboard with national-to-district drill-down.
- Provide a documented method for combining heterogeneous sources, early warnings, an API/export layer and a multilingual accessible interface.

For the SIH prototype, a narrow, honest pilot across selected sectors and districts is more credible than claiming complete national coverage.

3. Problem in everyday language

Imagine a district has enough training seats for 400 computer operators, but local employers advertise far fewer openings. At the same time, employers may struggle to find solar technicians or machine-maintenance workers. If planners only repeat last year’s targets, they may keep producing more of one skill while missing demand for another.

- Some occupations may have more trained people than local opportunities can absorb.
- Growing occupations may not have enough qualified people or training capacity.
- Data is split across surveys, job listings, worker registrations, training records and industry signals.
- Different sources use different occupation names, codes, time periods and geographic units.
- Planners need early evidence before the next training cycle and seat allocations are finalized.

This is not just a job-search portal, course directory, unemployment chart or generic chatbot. Its core is comparable demand and supply estimates, forecasts, explainable alerts and training-planning scenarios.

4. Existing ecosystem and research gap

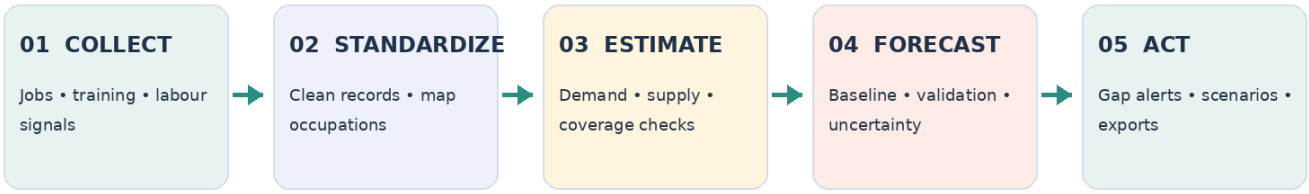
India already has useful components, but each serves a different purpose. The proposed system must connect evidence carefully rather than treat every source as interchangeable.

National Career Service (NCS)	Career/job services and job-related signals, subject to available access	A complete, deduplicated census of vacancies
PLFS / MoSPI / OGD	Labour-force context and statistical trends	Live district-level vacancy counts
e-Shram	Registration context for unorganised workers	That every registrant is currently available for a particular job
NCO-2015 / occupational classification	A common occupation classification and codes	A live hiring feed or qualification equivalence

NSQF / NCVET / NSDC resources	Qualification levels, occupational standards and training context	Current worker availability or guaranteed placement
Training-centre / scheme records	Seats, enrolment, completions, placements if definitions are available	The number of currently available qualified workers without further modelling

Research rule
For every dataset, record its owner, exact file/endpoint, fields, geography, date range, refresh frequency, access terms and limitations. If access has not been verified, label it as a proposed future integration.

5. Proposed solution: LMIS in one workflow



Every result should show: value + source + last updated + confidence/limitation.

The product has five stages: collect signals; standardize records; estimate demand and qualified supply; forecast the future; and help planners review alerts and scenarios. The stages must be auditable, and the dashboard must expose the evidence behind important results.

- Collect: approved APIs and validated CSV/XLSX uploads.
- Standardize: clean formats, normalize geography and map job titles to occupation codes.
- Estimate: create demand indicators and define the supply population clearly.
- Forecast: use baselines first, evaluate models over time and show uncertainty.
- Act: rank risks, compare training-seat scenarios and export evidence for human review.

6. Data strategy and source inventory

The data inventory is the first technical deliverable. Do not merge records just because they all mention employment. Each signal needs a definition, unit, geography, time period and reliability note.

Job demand signals	Posting ID, title, employer/industry if allowed, location, date, experience, source	Listings can be duplicated, stale, online-biased or not equal to hires
Labour-market statistics	Employment/unemployment measures, occupation/industry where available, geography, period	Survey estimates may not be current or granular enough for every district/occupation
Training capacity	Centre, course/trade, planned seats, year, district, provider	Seats are capacity, not workers already available
Training outcomes	Enrolment, completion, certification, placement and follow-up period	Definitions vary; placement is not the same as long-term employment
Worker registration/context	Occupation/skill fields and geography where lawfully available	Registration is not proof of current availability or job readiness
Industry signals	Employer surveys, expansion plans, sector forecasts, apprenticeship demand	May be partial, qualitative or difficult to compare

Recommended source status labels: (1) public real data, (2) authorized/imported data, (3) synthetic demonstration data, and (4) planned integration. Show the status in the UI and in exports.

Fallback plan: build a reliable import-and-validation pipeline first. If a source lacks a usable API or fine-grained data, use public aggregate context and clearly labelled synthetic records to demonstrate the end-to-end workflow.

7. Occupation mapping and normalization

The same work may be advertised using different job titles. For example, “Solar PV Installer” and “Solar Technician” may overlap, but should not be treated as identical automatically. A wrong mapping can distort demand, supply and forecasts.

- Clean titles: casing, punctuation, common abbreviations and obvious spelling variants.
- Use exact matches for verified aliases and known occupation codes first.
- Use semantic similarity only to suggest candidate mappings, not silently approve them.
- Display the proposed NCO/qualification mapping, confidence and reason.
- Send low-confidence or conflicting mappings to a human review queue.
- Keep the original title and source record for traceability; version the mapping rules.

A mapping record should store the raw title, canonical occupation, NCO code if mapped, method, confidence, reviewer status and mapping version.

8. Demand, supply and gap calculations



Gap = projected demand − projected qualified supply. These numbers are fictional, not official statistics.

Keep demand and supply separate until their definitions, geography, occupation and time periods line up.

Observed demand signal	A documented indicator built from available hiring/posting signals	Not necessarily total vacancies or total hiring
Projected demand	Demand estimate for a stated future period	Show model, horizon, last observation and uncertainty
Qualified supply	Defined estimate of workers plausibly qualified for the target occupation and place	Must not equal training seats or all registrations
Absolute gap	Projected demand – projected qualified supply	Positive = potential shortage; negative = potential oversupply
Gap rate	(Demand – Supply) / Demand × 100, when demand > 0	Show beside absolute gap; never use alone
Coverage / confidence	A transparent measure of data completeness and forecast reliability	Do not confuse this with a probability that a job will be filled

Illustrative example only: projected demand = 600; qualified supply = 420; gap = +180. These are fictional numbers, not live district statistics. A near-zero gap means estimates are close; it does not guarantee every person will find work.

Training seats are capacity. Enrolments are people who started training. Completions are people who finished. Placements are reported employment outcomes with a definition and follow-up period. These must remain separate fields.

9. Forecasting method and validation

Start with a simple baseline and keep a more complex model only if it improves results on held-out time periods. Deep learning is not automatically better, especially when history is short or data is sparse.

- Define the target, frequency, occupation/geography scope and forecast horizon.
- Build monthly or quarterly time series only where the data supports that frequency.
- Establish a naïve or moving-average baseline before testing more complex methods.
- Candidate methods may include exponential smoothing/ARIMA or tree-based models if enough relevant data exists.
- Use time-based holdouts or rolling-origin evaluation; do not randomly shuffle time-series rows.
- Report MAE or another suitable error measure against a baseline.
- Show uncertainty ranges only when the method and data support them.
- If evidence is too sparse, show “insufficient evidence for a reliable forecast.”

Each forecast should retain its target, geography, horizon, last observed date, model name/version, training window, evaluation metrics, uncertainty and data-source lineage.

10. Supply forecasting and seat scenarios

A simple supply model can be written as: next-period supply = current estimated supply + expected qualified entrants – expected exits + net movement. It is only a starting point; the team must define what each term means and avoid double counting.

A training-seat scenario should apply conversion assumptions. Not everyone offered a seat will enrol, complete training or enter the target occupation and geography.

Illustrative calculation
If an estimated shortage is 200 workers, expected completion is 85%, and 70% of completers enter the target occupation/geography, expected yield per seat = 0.85 × 0.70 = 0.595. A simple estimate is 200 ÷ 0.595 ≈ 336 seats. This ignores timing, existing pipeline, centre capacity, cost and uncertainty.

The scenario planner should let a user change assumptions, compare current plan versus proposed seats, show the expected effect on the gap, and clearly label the result as a scenario—not a recommended allocation automatically approved by government.

11. Alerts and recommendations

Shortage risk	Positive projected gap, demand trend, supply trend, coverage and uncertainty	Review whether additional seats, apprenticeships or employer-linked batches are justified
Oversupply risk	Negative projected gap, weak demand, training pipeline and coverage	Review new allocations and consider transferable skills or alternative roles
Emerging demand	Sustained growth in hiring signals or another reliable leading indicator	Review curriculum readiness and training capacity
Data warning	Stale source, sparse history, low mapping confidence or conflicting records	Avoid strong policy conclusions until the evidence is reviewed

Alert ranking should consider gap size, relative gap, persistence, urgency, evidence coverage and forecast reliability. Display the reasons behind the rank so a planner can challenge it.

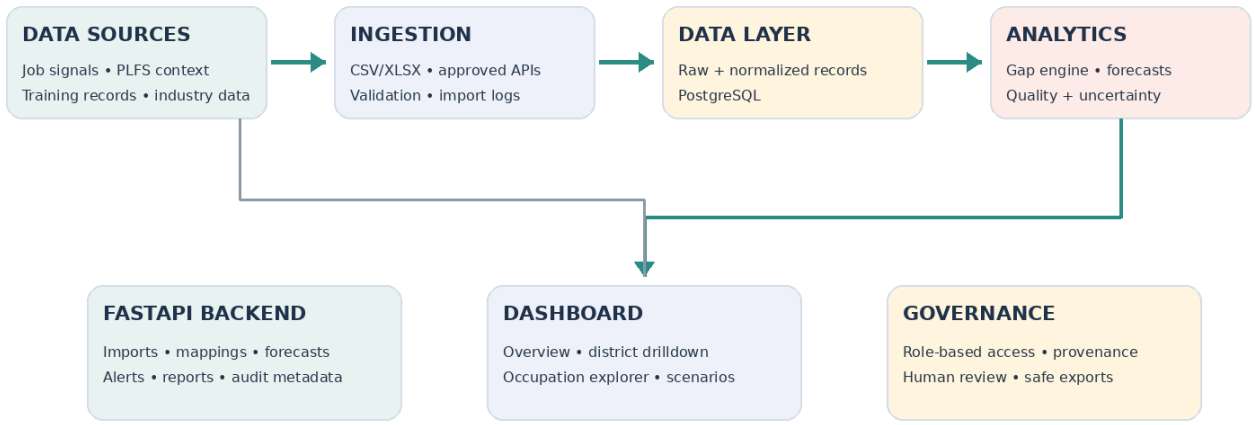
12. Dashboard and user experience

Overview	Demand/supply summary, occupations monitored, active alerts and data freshness	Choose state, district, sector and horizon
Geographic drill-down	Map or district list coloured by gap status	Select a district to view occupations and evidence

Occupation explorer	Canonical occupation, NCO mapping, qualification context and historical signals	Filter by sector, trade, district and qualification level where mapped
Forecast view	Observed history, demand/supply projections, gap and uncertainty	Compare supported forecast horizons
Early-warning inbox	Ranked shortage, oversupply, emerging-demand and data-quality alerts	Open the evidence and mark for review
Scenario planner	Current training plan versus proposed seat changes	Change assumptions and compare the estimated gap
Reports / data	Exports, source notes, import history, mapping reviews and API documentation	Download CSV/PDF and inspect provenance

Design principles: readable charts, plain-language labels, accessible contrast, keyboard navigation, responsive layout, multilingual-ready text, visible source dates and clear “real / imported / simulated” badges.

13. Technical approach and recommended stack



MVP principle: one modular application; no unnecessary microservices.

Live government connectors are future work until access, terms and endpoints are verified.

Frontend	React + TypeScript; Tailwind or consistent CSS	Reusable components and reliable interactions
Charts / maps	Recharts or ECharts; Leaflet if map is needed	Trend, comparison and geographic drill-down
Backend API	Python + FastAPI	Clear endpoints and automatic OpenAPI docs
Data handling	Pandas	CSV/XLSX validation and feature preparation
Forecasting	Statsmodels + scikit-learn where justified	Baselines and candidate statistical/ML models
Database	PostgreSQL; PostGIS only if needed	Structured records and optional geographic queries
Scheduled work	Cron / simple scheduled job initially	Enough for periodic refresh in an MVP
Deployment	One deployable application and managed database where possible	Reduces operational complexity

Prefer a modular monolith over microservices for SIH. Separate ingestion, normalization, analytics, API and dashboard in code, but keep deployment simple. Do not add AI components merely to make the project sound more advanced.

14. Suggested data model and API

data_source	Source name, owner, access method, terms, refresh frequency, limitation
ingestion_run	Source, timestamp, status, rows accepted/rejected, error summary
raw_observation	Original record/payload, source ID, observed date
occupation / alias	NCO code, canonical title, sector; aliases, match method, confidence, review status
geography	Standard state/district identifiers and validity dates
demand_observation	Occupation, geography, period, signal value, source and method
training_capacity / outcome	Centre, trade, seats, enrolments, completions, placements and definitions
supply_estimate	Occupation, geography, period, estimate, method and uncertainty
forecast_run / value	Model version, target, horizon, metrics, estimate, lower/upper bounds
alert	Type, severity, evidence, status, creation date and review history

GET /api/v1/overview	Summary counts and freshness for selected filters
GET /api/v1/occupations	Search occupations and mappings
GET /api/v1/gaps	Demand, supply, gap, period and evidence metadata
GET /api/v1/forecasts	Forecast values, horizon, model version and uncertainty
GET /api/v1/alerts	Early-warning alerts with evidence and review status
POST /api/v1/imports	Validated CSV/XLSX import with an import audit record
GET /api/v1/reports/{id}/export	Download tabular analysis or report

API responses should include source/status metadata and should not return a naked number without its period, geography, occupation definition and caveat.

15. Security, privacy, fairness and governance

- Collect only the data needed for planning; avoid personally identifiable worker-level data in the MVP unless strictly necessary and authorized.
- Use role-based permissions for data import, mapping approval, report viewing and administration.
- Keep an audit trail for imports, mapping edits, model runs, exports and user actions.
- Document source permissions, terms, consent/legal basis and retention requirements before any integration.
- Do not expose sensitive individual records in dashboards or exports.
- Test for geographic and occupational coverage gaps; online job postings can overrepresent formal/digital hiring.
- Show missingness and uncertainty instead of presenting a biased estimate as complete truth.
- Keep recommendations as human-reviewed decision support; never automatically allocate seats or close training centres.

16. Feasibility, viability and risks

Technical	CSV/XLSX import, normalization, occupation mapping, gap calculation, baseline forecast, dashboard and exports	Stable production connectors, high-scale ingestion, production identity/security
Data	Public context data and authorized files where available; labelled synthetic data to exercise full flow	Verified district/occupation-level histories and rights to use all desired sources
Operational	Pilot in a small number of sectors/districts; assign data and connector owners	Long-term source ownership, maintenance budget, security reviews and training
Policy	Evidence and scenarios for authorized planners to review	Official target-setting process, approved access and governance

Data access weaker than expected	Start with source audit and validated file imports; never claim an untested live connector
Duplicate or stale postings	Deduplicate with documented rules, retain timestamps and show uncertainty
Misleading supply estimate	Define the target population and do not equate registrations, certificates or seats with available workers
Short history / weak forecasts	Baseline, time-based validation, uncertainty labels and insufficient-evidence state
Occupation mismatch	Official reference codes, confidence thresholds and human review
Overconfident policy recommendation	Explain assumptions and require authorized human review
Scope creep	Protect the end-to-end MVP; defer optional integrations and advanced models

17. MVP scope and build roadmap

1. Verify	Source inventory, pilot sector/district choice and definitions	Each dataset has fields, access method, date range and limitations recorded
2. Foundation	Repository, database, validated import and sample data	A clean import produces normalized records and an audit entry
3. Core logic	Occupation mapping, demand/supply definitions and gap calculation	Known test cases return expected values
4. Forecast	Baseline and time-based evaluation if history supports it	Outputs include horizon, last observation, metrics and caveats
5. Dashboard	Overview, occupation explorer, district drill-down and forecast views	UI reads from working API endpoints
6. Decisions	Alerts, seat scenarios and exports	Planner can test a scenario and export assumptions/results
7. Harden	QA, setup guide, demo rehearsal and documentation	Demo runs from a fresh setup without manual patching

Recommended work split: research/data lead (source inventory); backend lead (schema/import/API); analytics lead (gap and forecast evaluation); frontend lead (dashboard); QA/integration lead (tests and demo); UX/presentation role (visual system and judge narrative). People can hold multiple roles, but every deliverable needs one owner.

18. Test and evaluation checklist

- Test malformed files, missing columns, duplicate rows, invalid dates and inconsistent geography codes.
- Test occupation mapping confidence and human review paths.
- Verify gap arithmetic, zero-demand edge cases and sign interpretation.
- Verify forecasts use only information available before each test period.
- Compare forecast error with a simple baseline and document results.
- Verify missing or stale data triggers warnings rather than false precision.
- Test role permissions, audit events and export filtering.
- Check accessibility, responsiveness, loading/empty/error states and multilingual layout.
- Run the complete demo from a clean environment and retain a short evidence log.

19. What to show judges: demo narrative

Recommended 3–5 minute demo: start with the planning problem; choose a pilot sector and district; show source freshness and whether records are real or simulated; open an occupation; show demand, supply, gap and forecast with evidence; open an alert; change training seats in the scenario planner; export a report with assumptions and limitations.

Proof over promises
Show the working path end to end. If a live government API is not available and tested, describe it as a planned connector, not a completed integration.

20. Judge questions and defensible answers

How is this different from a normal dashboard?	It connects normalization, demand/supply estimation, forecasting, explainable alerts and training scenarios in one workflow.
Where is the AI?	Semantic occupation matching can suggest mappings and forecasting models can estimate trends when data supports them. Gap arithmetic remains deterministic.
How do you know actual supply?	We estimate a defined qualified-worker pool from available evidence and disclose uncertainty; we do not claim exact availability if data cannot establish it.
How do you avoid duplicate vacancies?	Use source IDs and matching rules; unresolved duplicates remain uncertain rather than counted as certain unique vacancies.
What if official APIs are unavailable?	The MVP supports validated imports and documents future connectors. No integration is called live unless tested.
How are forecasts validated?	Time-based holdouts or rolling-origin evaluation, comparison with a baseline, error metrics and uncertainty.
Can the system decide training seats?	It provides evidence and scenarios; authorized planners retain the decision.
What is real versus simulated?	Every source is labelled. Synthetic data demonstrates workflow only and is never presented as official statistics.

21. Impact and benefits

- Better alignment between training capacity and observed/emerging employer demand.
- Earlier warning of possible shortages, oversupply and changing occupation demand.
- More transparent planning through source lineage, model notes and visible uncertainty.
- Comparable evidence across selected sectors and districts, where data supports comparison.
- Faster exploration of training-seat scenarios before target-setting discussions.
- Reusable ingestion, normalization and export patterns for future approved sources.

These are intended benefits, not already measured outcomes. A pilot should measure forecast error, source coverage, time to produce an analysis, number of records needing review, and whether planners find the scenarios useful.

22. Critical evaluation: what could make this fail?

- Data availability: the ideal data may not be public, granular or machine-readable.
- Representativeness: online postings may miss informal hiring and overcount reposted roles.
- Supply ambiguity: registrations, training seats, completions and current availability are different concepts.
- Forecast reliability: sparse time series can make forecasts unstable.
- Mapping errors: similar titles may represent different tasks or qualification levels.
- Policy overreach: a forecast alone cannot account for costs, local development plans, worker preferences or migration.

The strongest defence is disciplined scope, transparent definitions, baseline comparison, human review and a demo that clearly distinguishes verified evidence from assumptions.

23. Research references and links

Use official or primary sources first. Confirm the specific dataset, endpoint, terms and update schedule before implementation; a portal landing page does not itself guarantee API access.

- **National Career Service (NCS)**: https://ncs.gov.in/
- **NCS About**: https://ncs.gov.in/about-us
- **Directorate General of Employment — NCO-2015**: https://www.dge.gov.in/nco-2015
- **Searchable NCO resource**: https://dge.gov.in/nat
- **MoSPI — PLFS Annual Report 2023–24**: https://mospi.gov.in/sites/default/files/publication_reports/AnnualReport_PLFS2023-24L2.pdf

• **Open Government Data — PLFS datasets**: https://www.data.gov.in/keywords/plfs

• **e-Shram portal**: https://www.eshram.gov.in/e-shram-portal

• **NCVET — National Skills Qualification Framework**: https://ncvet.gov.in/national-skills-qualification-framework/

• **NCVET — NSQF notification**: https://ncvet.gov.in/national-skills-qualification-framework/nsqf-notification/

• **NSDC — National Occupational Standards and model curriculum**: https://nsdcindia.org/national-occupational-standards-and-model-curriculum

• **MSDE Annual Report 2021–22 (historical context)**: https://msde.gov.in/sites/default/files/2022-06/Annual%20Report%202021-22%20Eng.pdf

Reference deck pattern used for presentation planning: the uploaded six-page SIH deck is organized as (1) cover/problem statement, (2) product name + current problems vs solution, (3) technical approach/architecture + stack, (4) workflow, (5) feasibility & viability + risks/controls, (6) impact & benefits + research references. The new LMIS presentation should preserve this sequence while replacing GovSync-specific content with LMIS-specific content.

24. Presentation blueprint matching the supplied deck

1. SIH cover	Official PS ID/title/theme/category and registered team details (confirm exact team metadata)	Clean title page; no unsupported claims
2. Product overview	Proposed product name; current problems on left; solution capabilities on right	Problem → capability two-column layout
3. Technical approach	Tech stack on left; end-to-end architecture and major services on right/bottom	Architecture diagram with ingestion, normalization, analytics, API and dashboard
4. Workflow	Numbered user journey from data import/source to forecast, alert, scenario and export	Flowchart with evidence checkpoints
5. Feasibility & viability	Prototype status, technical feasibility, operational plan, risks and controls	Three-column layout: feasible, risks/controls, rollout viability
6. Impact & benefits + references	Expected impacts, benefits and 4–5 credible primary references	Two-column impact list and numbered source cards

Visual style to preserve from the reference: short headings, dense but scannable two-column layouts, a central architecture diagram, a numbered workflow slide, a three-column feasibility/risks slide and a final impact/references slide. Do not copy GovSync's content or its technology stack blindly; only reuse the presentation structure.

25. AI handoff instructions: how another AI should use this R&D

Copy the following instruction together with this Master R&D document when asking an AI to create the new SIH presentation:

Reusable prompt
You are preparing a Smart India Hackathon 2026 presentation for the problem statement “AI-Enabled Labour Market Intelligence and Skill Demand-Supply Forecasting Engine.” Read the entire attached Master R&D before creating slides. Use the uploaded reference deck only as a structural/style guide: reproduce its six-part sequence—(1) SIH cover/PS metadata, (2) product overview with current problems vs solution, (3) technical approach with stack and architecture, (4) numbered workflow, (5) feasibility/viability with risks and controls, and (6) impact/benefits plus research references. Replace all GovSync-specific concepts with LMIS content. Keep text concise enough for slides, but preserve technically meaningful details. Use only the stack actually selected by the team. Do not invent live integrations, validated nationwide forecasts, pilot results, official team metadata or measured impact. Label synthetic data and assumptions clearly. Cite credible official sources, use clean readable diagrams, and keep the output editable. If a required fact is missing, flag it rather than guessing.

Before generating the presentation, the team should fill in: product name; exact PS ID from the official portal; team ID/name; pilot sector(s)/state(s)/district(s); selected tech stack; real datasets actually obtained; features actually working; prototype URL (if any); and team member roles.

26. Final recommendation

Build the smallest end-to-end system that can prove the core idea: validated data import → occupation normalization → clearly defined demand/supply indicators → gap calculation → baseline forecast if data supports it → evidence-backed alert → seat scenario → export. A polished, honest pilot is stronger than a wide feature list that cannot be demonstrated.

Treat the Master R&D as the single source of truth. Update it whenever the team verifies a data source, changes the architecture, chooses a pilot scope or completes a feature. Keep a separate evidence log for what is implemented versus planned.